PRACTICAL NO 7: Configure IOS Intrusion Prevention System (IPS) Using the CLI

The Cisco IOS IPS acts as an in-line intrusion prevention sensor, watching packets and sessions as they flow through the router and scanning each packet to match any of the Cisco IOS IPS signatures. When it detects suspicious activity, it responds before network security can be compromised and logs the event through Cisco IOS syslog messages or Security Device Event Exchange (SDEE). The network administrator can configure Cisco IOS IPS to choose the appropriate response to various threats. The Signature Event Action Processor (SEAP) can dynamically control actions that are to be taken by a signature event on the basis of parameters such as fidelity, severity, or target value rating. These parameters have default values but can also be configured through CLI. When packets in a session match a signature, Cisco IOS IPS can take any of the following actions, as appropriate:

1) Send an alarm to a syslog server or a centralized management interface
2) Drop the packet
3) Reset the connection
4) Deny traffic from the source IP address of the attacker for a specified amount of time
5) Deny traffic on the connection for which the signature was seen for a specified amount of time

Cisco developed its Cisco IOS software-based intrusion-prevention capabilities and Cisco IOS Firewall with flexibility in mind, so that individual signatures could be disabled in case of false positives. Generally, it is preferable to enable both the firewall and Cisco IOS IPS to support network security policies. However, each of these features may be enabled independently and on different router interfaces.

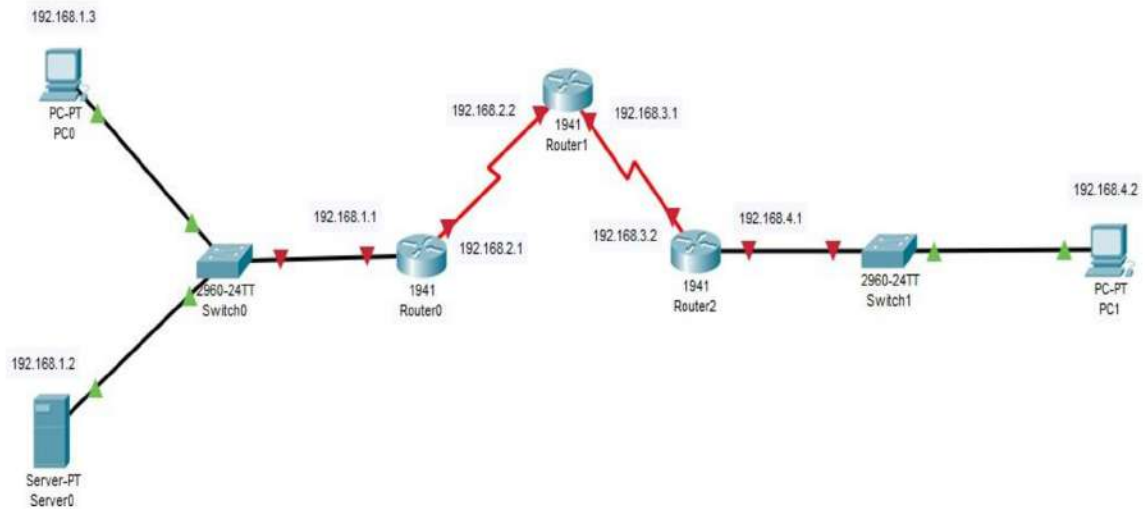
Signatures:

A signature is a set of rules that an IDS and an IPS use to detect typical intrusive activity, such as DoS attacks. We can easily install signatures using IDS and IPS management software such as Cisco IDM. Sensors enables us to modify existing signatures and define new ones.
As sensors scan network packets, they use signatures to detect known attacks and respond with predefined actions. A malicious packet flow has a specific type of activity and signature, and an IDS or IPS sensor examines the data flow using many different signatures. When an IDS or IPS sensor matches a signature with a data flow, the sensor takes action, such as logging the event or sending an alarm to IDS or IPS management software, such as the Cisco SDM

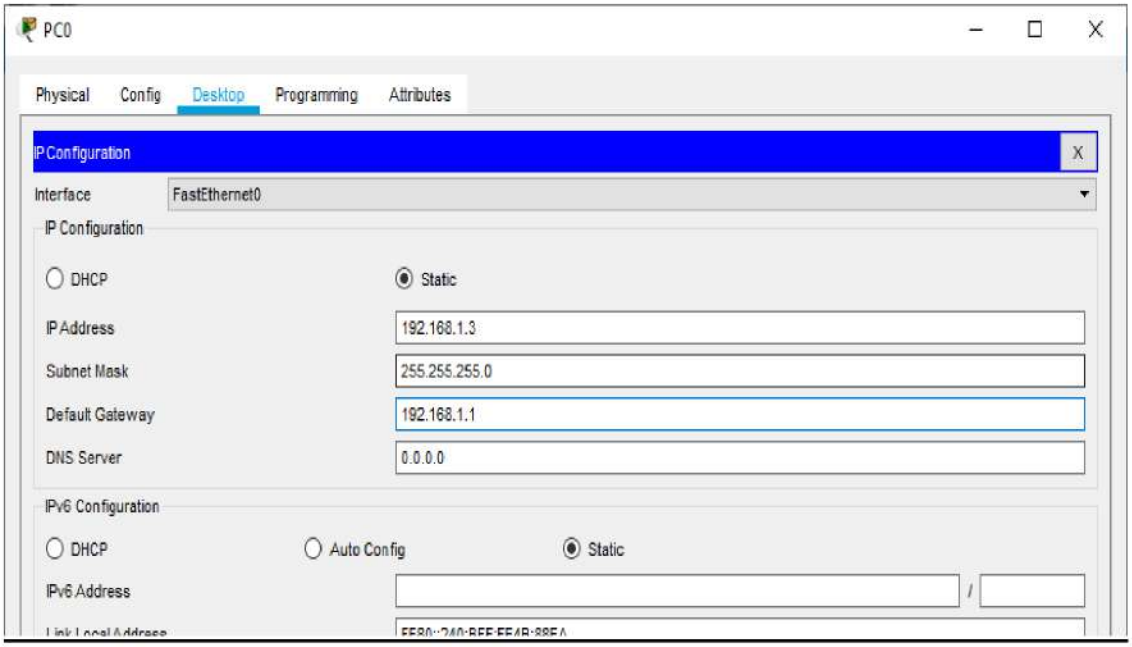
We define some of the commands which will be used while configuring the Router for IPS

Commands	Function	Example
ip ips signature-category	Enters IPS category configuration mode.	Router(config)# ip ips signature-category
category	Specifies that all categories (and all signatures) are retired in the following step and enters IPS category action configuration mode Specifies the basic category (and a set of signatures) that are to be "unretired" in the following step.	Router(config-ips-category)# category all **Example:** Router(config-ips-category)# category ios_ips basic
retired {**true** | **false**}	Specifies that the device should retire all categories (and all signatures). **true** --Retires all signatures within a given category. **false** --"Unretires" all signatures within a given category.	Router(config-ips-category-action)# retired true
mkdir flash:/ips5	Create a directory for which Cisco IOS IPS saves signature information.	**Example:** Device# mkdir flash:/ips5
ip ips name *ips-name*		**Example:** Device(config)# ip ips name myips
ip ips *ips-name* {**in** | **out**}	Applies an IPS rule at an interface and automatically loads the signatures and builds the signature engines.	**Example:** Device(config-if)# ip ips MYIPS in

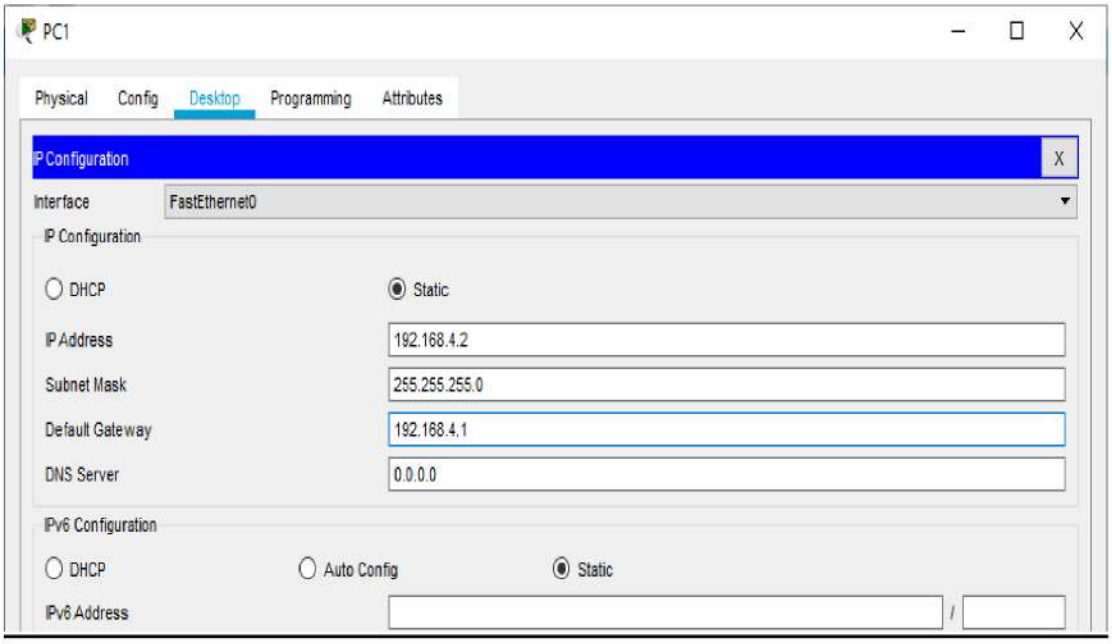
We us the following topology for the present case



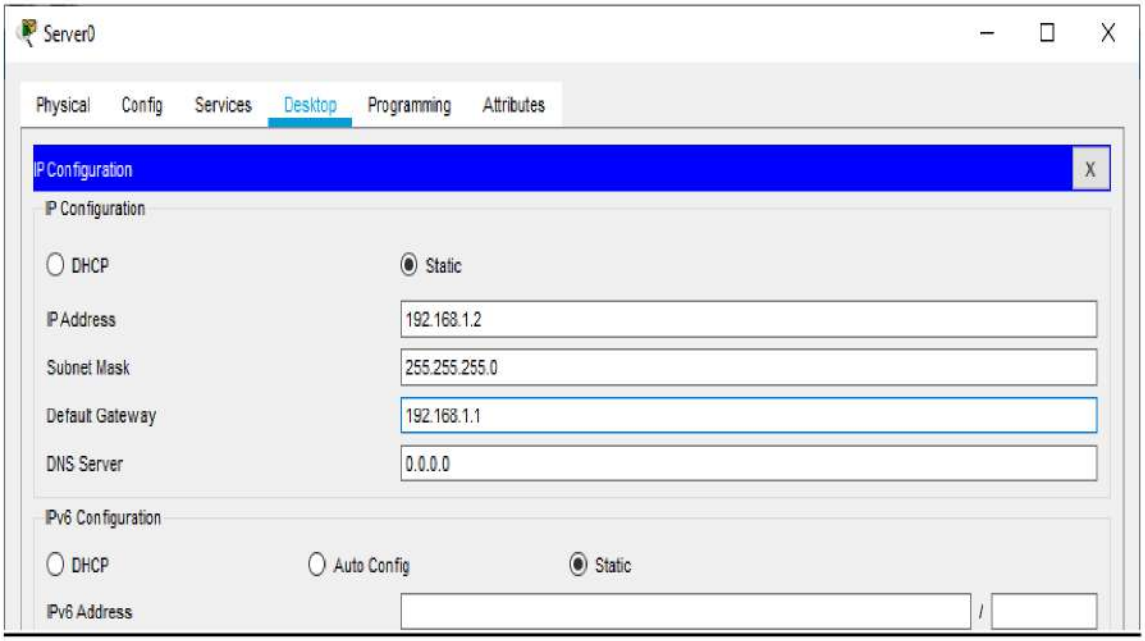
Configuring PC0



Configuring PC1

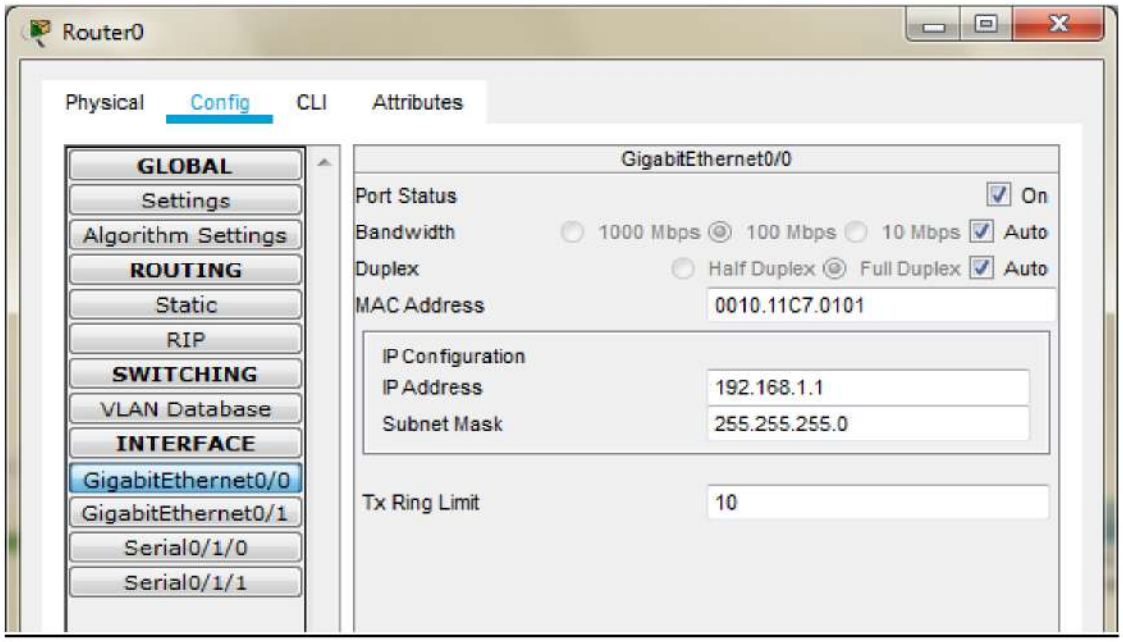


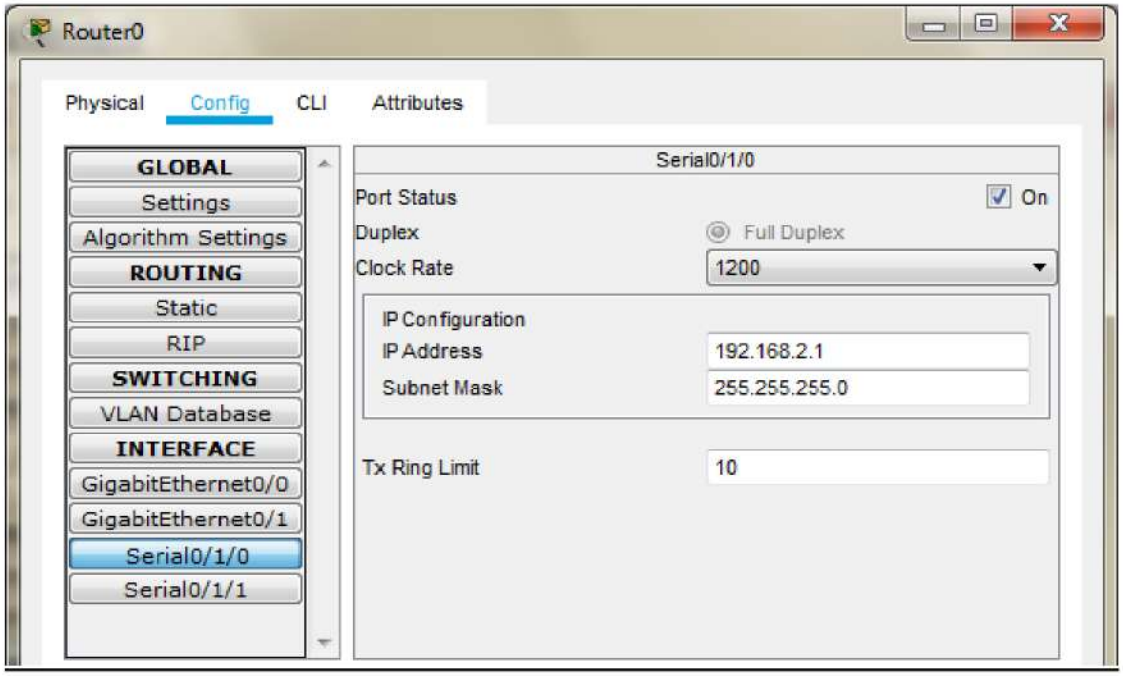
Configuring SERVER0



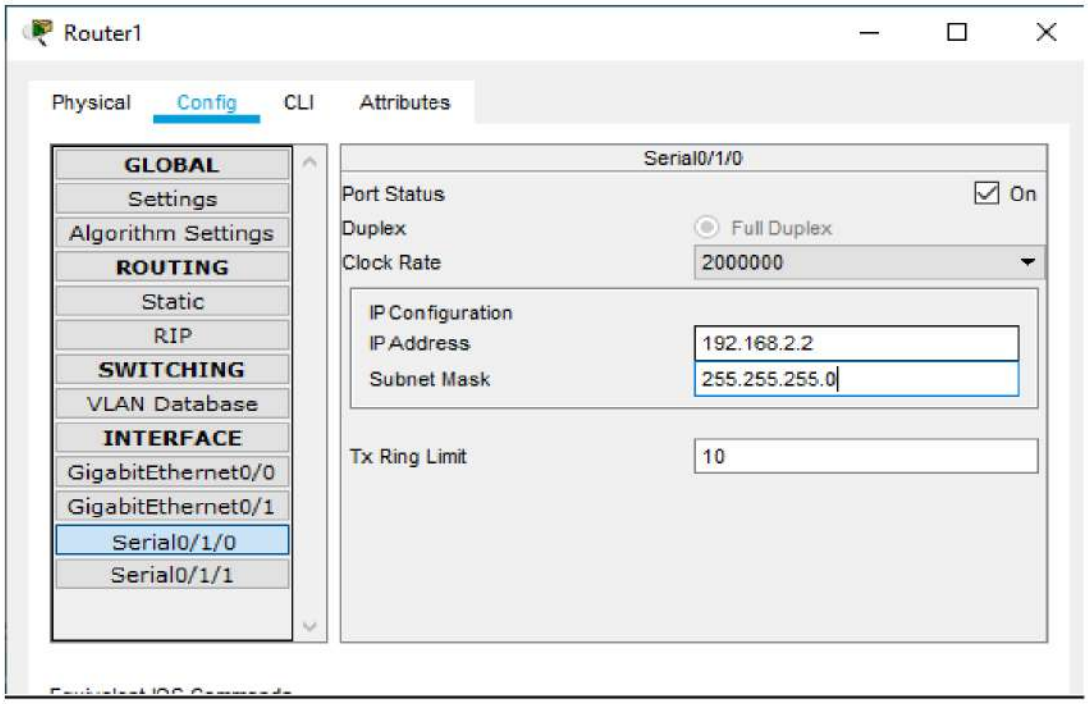
Serial Interface must be added in each Router before configuring it

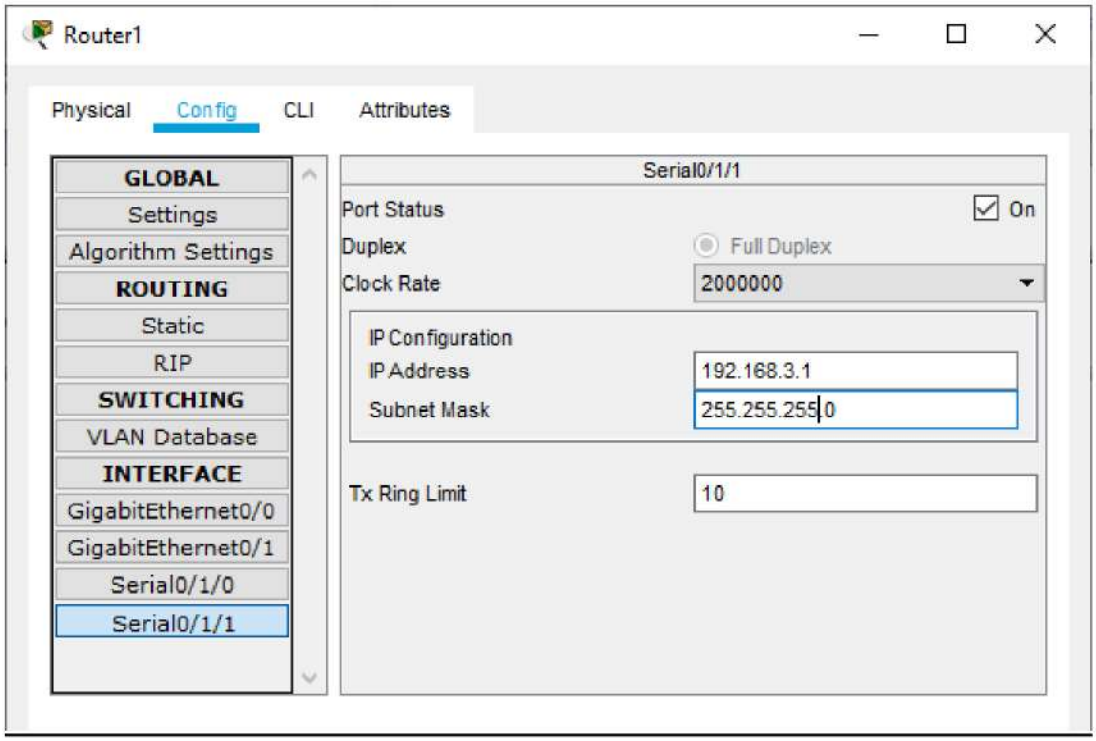
Configuring Router0



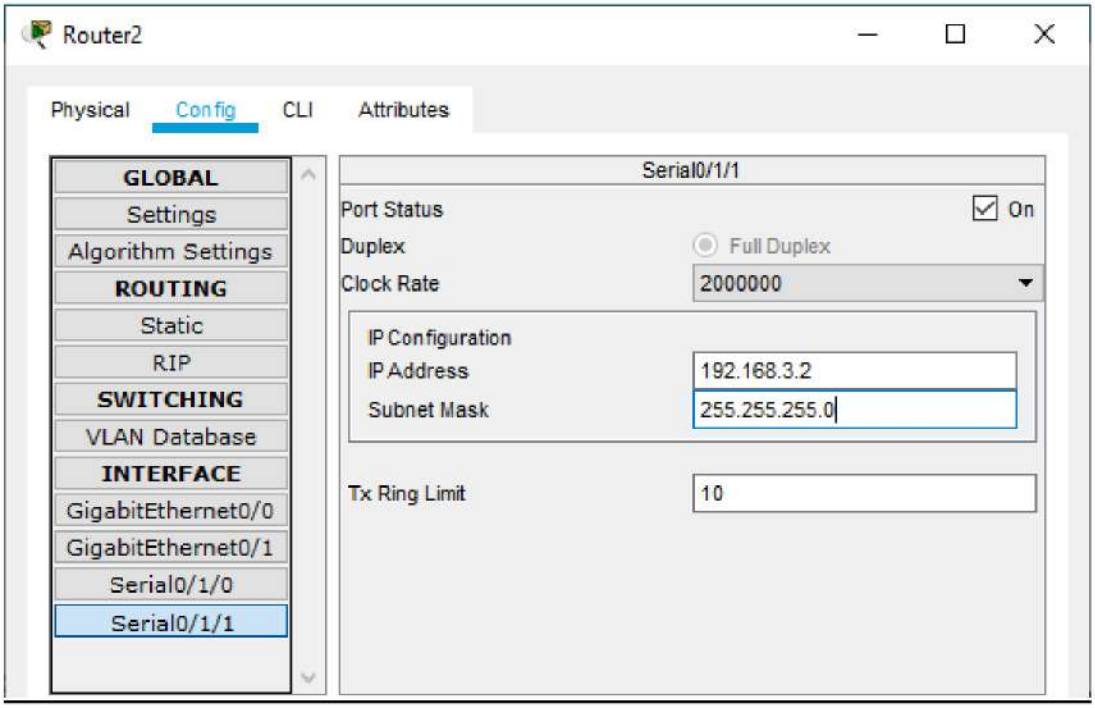


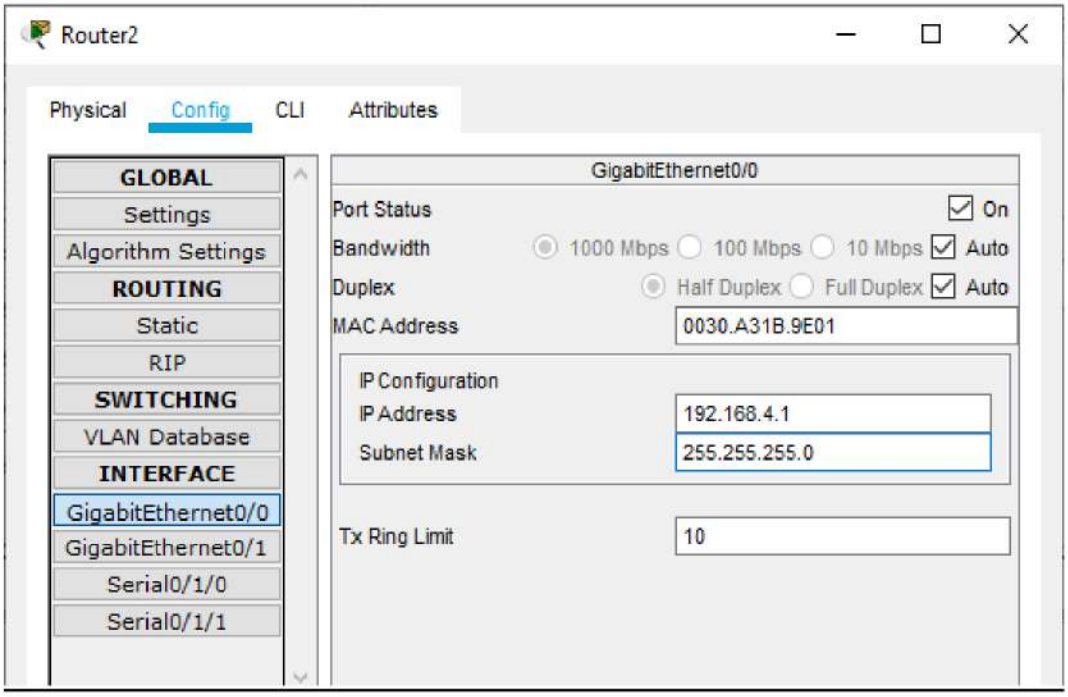
Configuring Router1





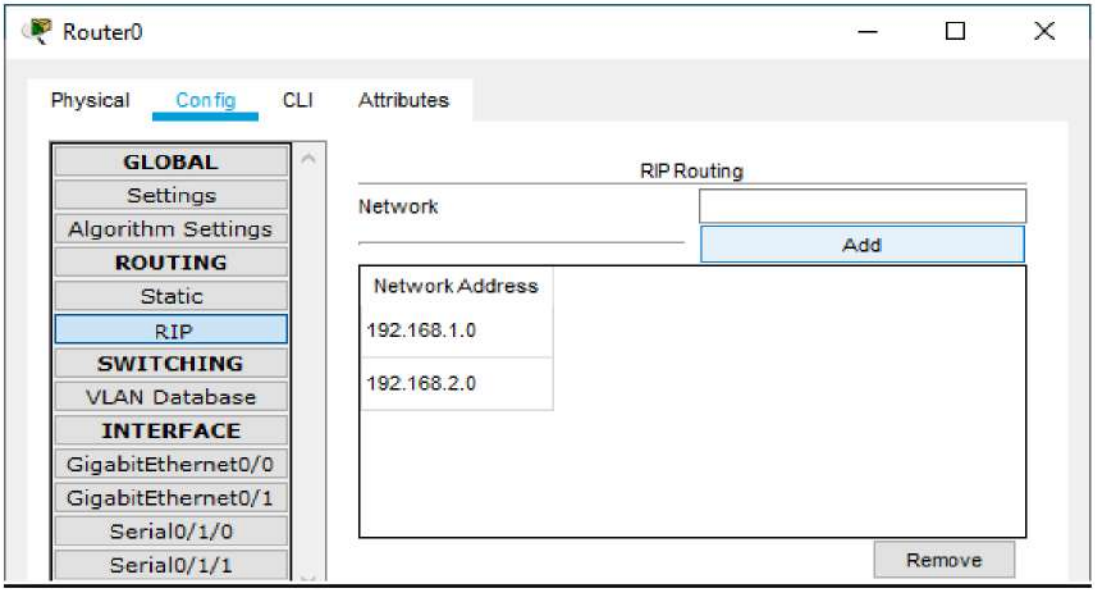
Configuring Router2



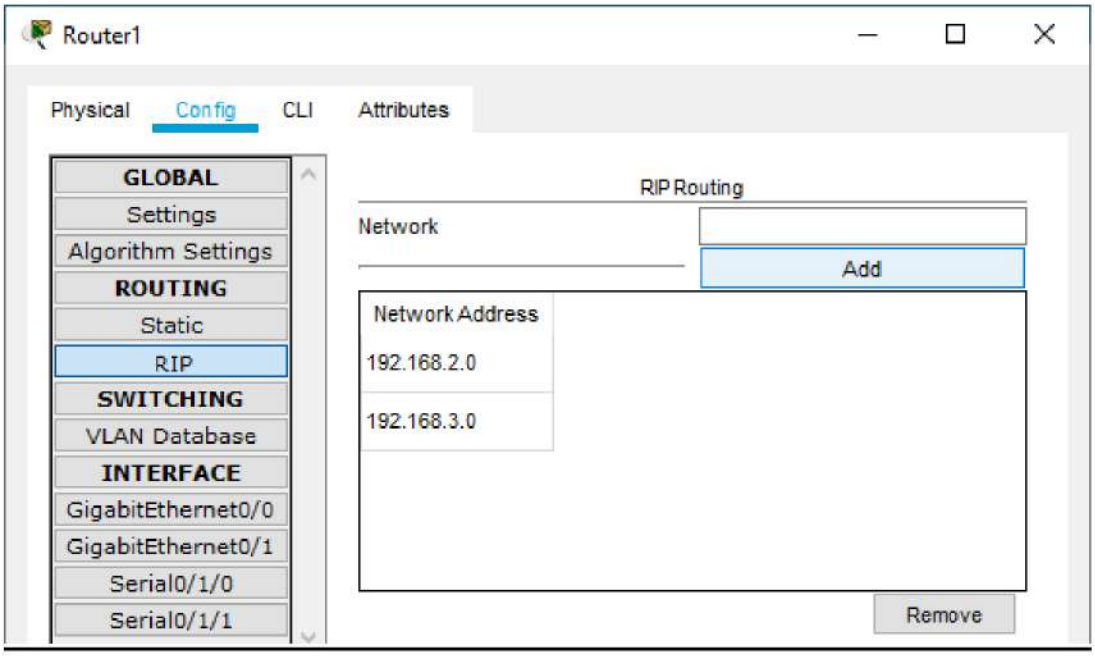


We need to set the Routing table in all the Routers so that each node could send and receive packets from others (RIP is set in all the Routers as follows)

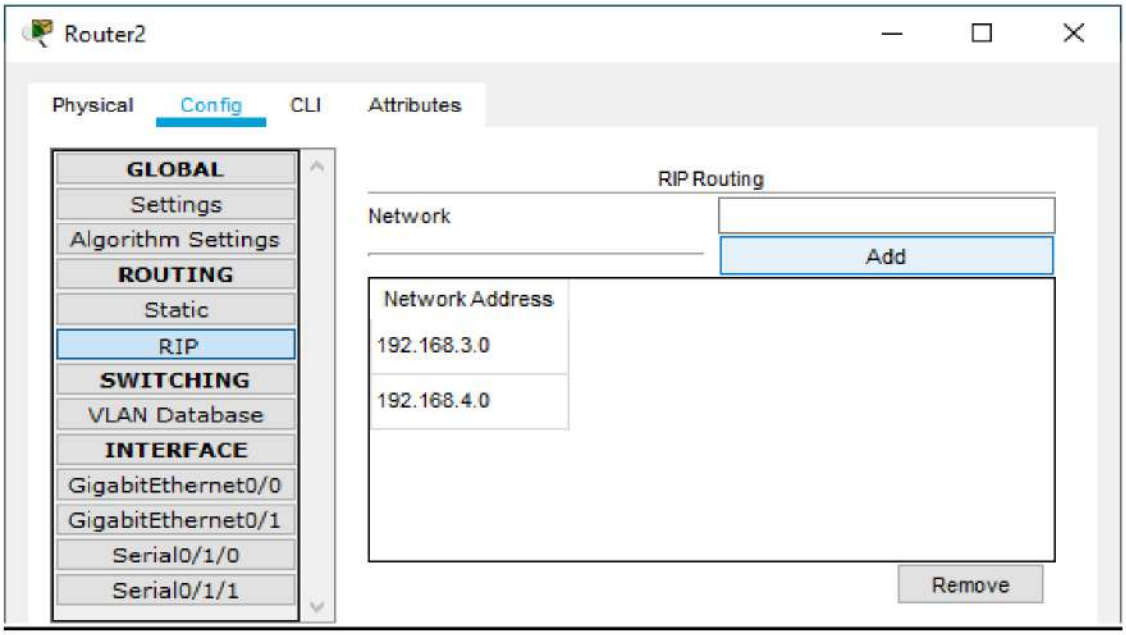
Router0



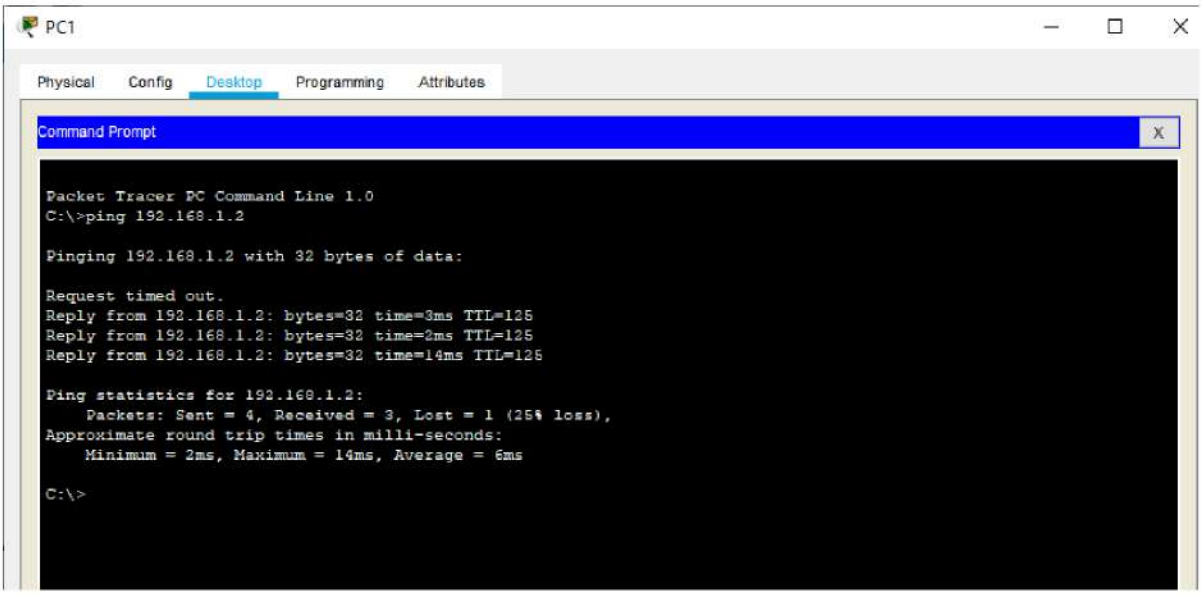
Router1



Router2



Now we can check the connectivity by sending ping commands from any node to any other node



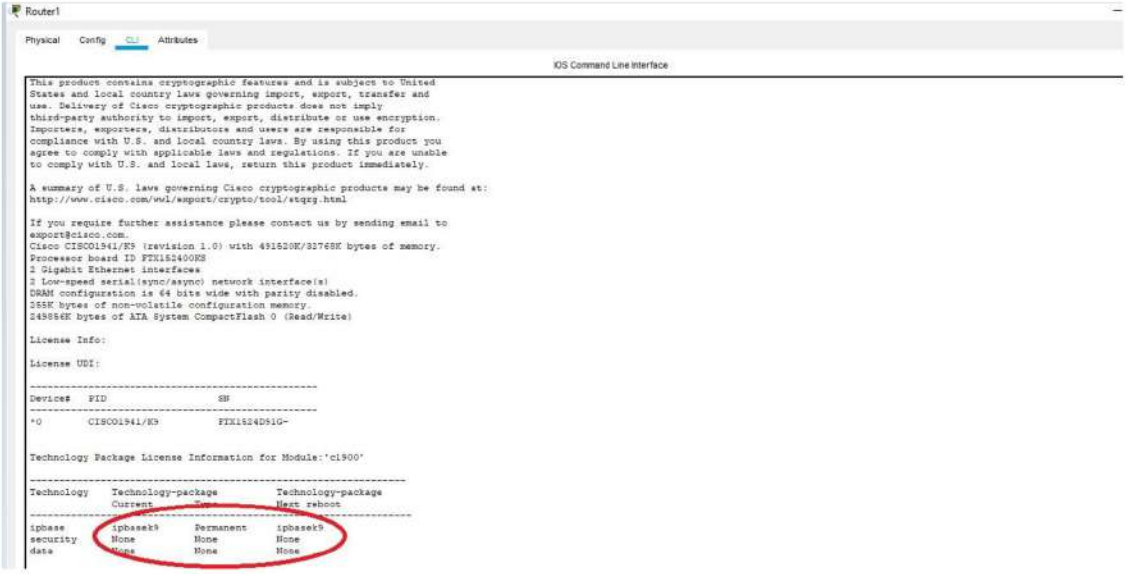
So we conclude that the connectivity has been established

PART1: Enable the IOS IPS (on Router1)

Type the following command in the CLI mode of Router1

Router#show version

We will get a message informing whether the security Package is enabled or not



As seen above the security package is not enabled, to enable the security feature, type the following command in Router1

Router(config)#license boot module c1900 technology-package securityk9
Router(config)#exit
Router#
Router#reload

Router>enable
Router#
Router#show version

We will get a message informing whether the security package is enabled or not

```
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX1524D91G
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:

License UDI:

-------------------------------------------------
Device#   PID                    SN
-------------------------------------------------
*0        CISCO1941/K9           FTX1524D91G-


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102
```

As seen above now the security package has been enabled

Now type the following commands in the CLI mode of Router1

```
Router#
Router#
Router#clock set 11:47:56 MARCH 3 2020

Router#mkdir smile
Router#configure terminal
Router(config)#ip ips config location flash:smile
Router(config)#ip ips name iosips
Router(config)#ip ips notify log
Router(config)#ip ips signature-category
Router(config-ips-category)#category all
Router(config-ips-category-action)#retired true
Router(config-ips-category-action)#exit

Router(config-ips-category)#category ios_ips basic
Router(config-ips-category-action)#retired false
Router(config-ips-category-action)#exit
Router(config-ips-category)#exit

Router(config)#interface Serial0/1/0
Router(config-if)#ip ips iosips out
Router(config-if)#
Router(config)#
```

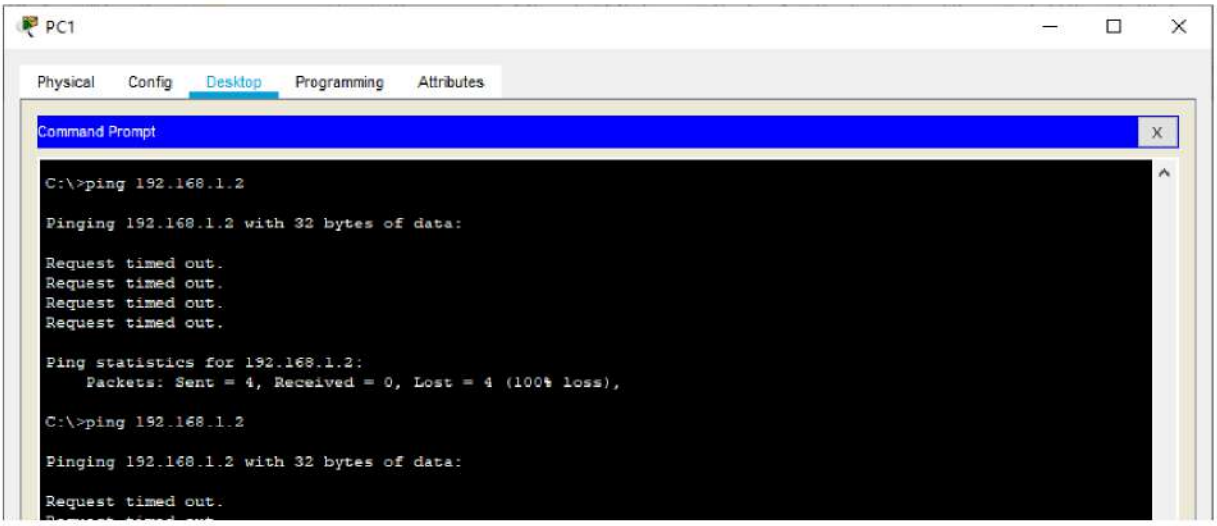
Part 2: Modify the Signature

Type the following commands in the CLI mode of Router1

```
Router(config)#
Router(config)#ip ips signature-definition
Router(config-sigdef)#signature 2004 0
Router(config-sigdef-sig)#status
Router(config-sigdef-sig-status)#retired false
Router(config-sigdef-sig-status)#enabled true
Router(config-sigdef-sig-status)#exit
Router(config-sigdef-sig)#engine
Router(config-sigdef-sig-engine)#event-action produce-alert
Router(config-sigdef-sig-engine)#event-action deny-packet-inline
Router(config-sigdef-sig-engine)#exit
Router(config-sigdef-sig)#exit
Router(config-sigdef)#exit
Router(config)#
```

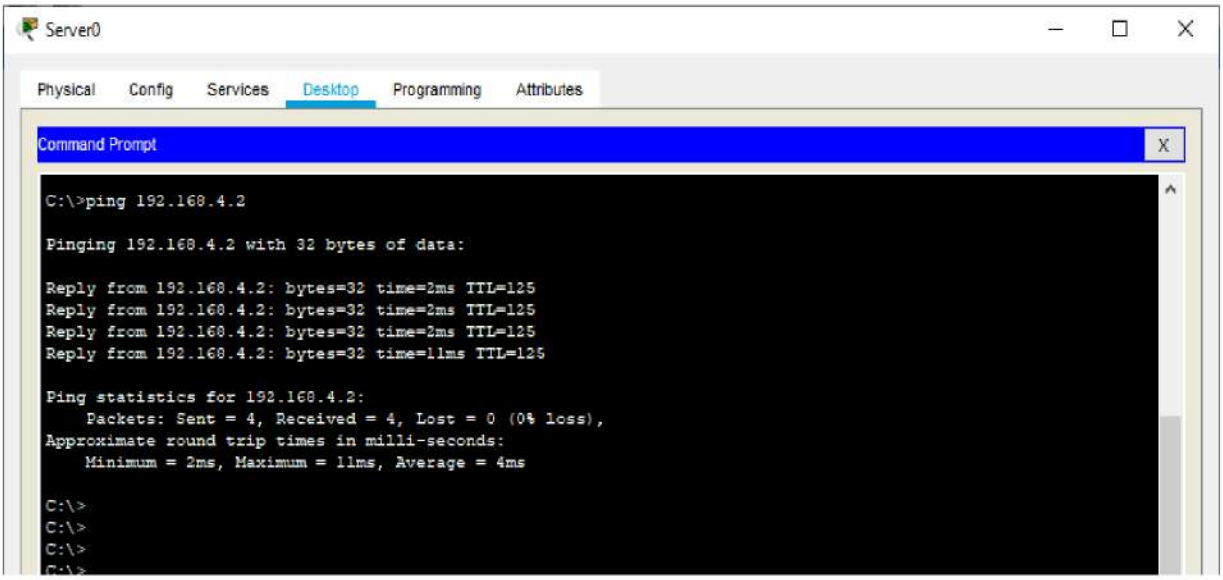
Now we need to verify the above IPS configuration, we do it first by pinging PC1 to SERVER and then from SERVER to PC1

PC1 to SERVER

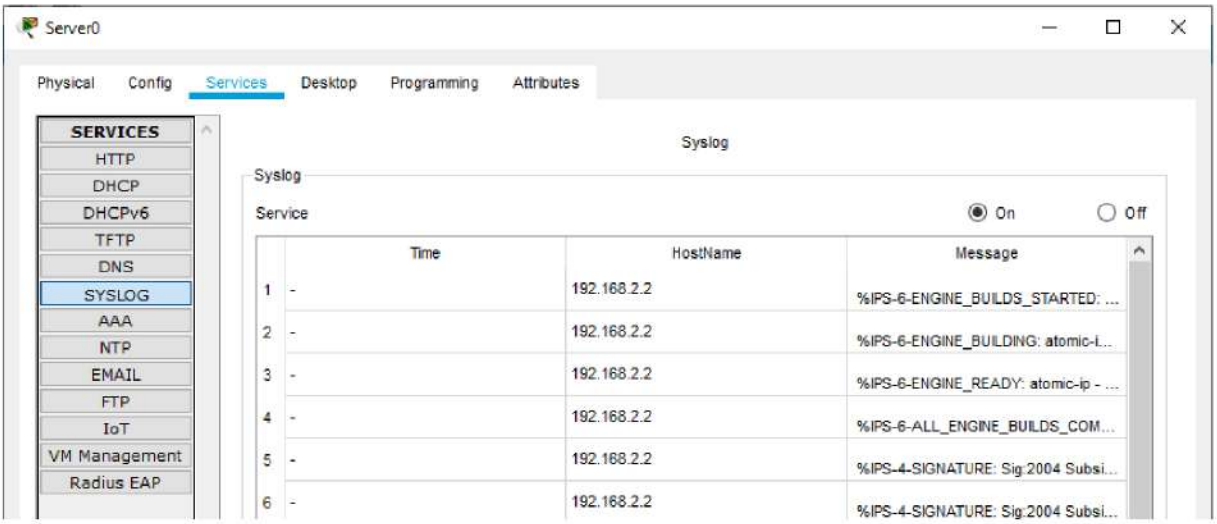


The ping FAILS

SERVER to PC1



Also we can observe the Syslog service in the SERVER to check the log activities



Hence we set the IPS and also verified it on Router1